

ARPITA ROY

Boston, MA | roy.arp@northeastern.edu | 857-961-7218 | [linkedin.com/in/arpita-roy-733132273/](https://www.linkedin.com/in/arpita-roy-733132273/)

EDUCATION

Northeastern University, Boston, MA, USA

December 2025(Expected)

Master of Science in Engineering Management

GPA: 3.53

RELEVANT COURSES: Probability and Statistics, Deterministic Operations Research, Engineering Project Management, Product Development for Engineers, Economic Decision Making, Supply Chain Engineering, Lean Concepts and Applications

SKILLS

Analytics : Power BI, Tableau, Excel (VLOOKUP, Pivot Tables), MySQL

Programming & Simulation : C, VBScript, AMPL, LINGO, Anylogistix

Certifications : Lean Green Belt

Lean & Quality Tools : Six Sigma, DMAIC, Kanban, FMEA, Root Cause Analysis, VSM, 5S, Kaizen

PROFESSIONAL EXPERIENCE

XL Batteries, Marlborough, MA, USA

Jan 2025 to Aug 2025

Prototype Engineer Co-op

- Spearheaded targeted leak testing protocols on organic flow battery prototypes, **increasing operational lifespan** and **reducing electrolyte leakage** by 20% through improved sealing techniques and material selection.
- Fabricated 100+ organic flow battery prototypes using CNC routers and power tools while adhering to strict tolerance requirements, contributing to successful completion of initial design verification testing.
- Collaborated with cross-functional teams to identify continuous process improvement opportunities leveraging **Lean Six Sigma**, **5S** and **Kaizen** to reduce **waste** and **workplace clutter**, improving safety and workflow.
- Drafted manufacturing SOPs to specifications, maintaining strict compliance with all specified requirements.

GST & Central Excise (Government of India, Indirect Taxation), Kolkata, WB, India

Jun 2016 to Aug 2023

Intelligence Officer/ Executive Manager

- Analyzed data and prepared **dashboards** and **analytics** using **SQL**, **Excel**, **Power BI**, and **Tableau**, to create insightful reports and enhance existing reporting requests for management.
- Collaborated with stakeholders to optimize **data flows**, **data management**, and tracking processes, resulting in a 10 % reduction in data processing time and a 15% improvement in data accuracy.
- Developed a deep understanding of business domains, identified areas for **process improvement** and generated new data-driven insights, enhancing efficiency of existing business processes by 12%.
- Demonstrated strong project skills to **prioritize tasks**, manage workload, address challenges, and implement **mitigation strategies**, resulting in a 10 % increase in productivity.

Tata Consultancy Services Limited, Kolkata, WB, India

Mar 2013 to Jun 2016

Systems Engineer

- Scripted automation coding (**VB Scripts**) in **QTP**, enhancing overall project efficiency by **15%**.
- Delivered quality assurance solutions for Staples Inc.'s **Order Management System** and delivered dedicated technical solutions to customers, enhancing user satisfaction by 18%.

PROJECT EXPERIENCE

Northeastern University, Boston, MA, USA

Sep 2023 to Apr 2024

Printer Locator for Northeastern:

- Conceptualized a **web-based application** and implemented new technologies to track the nearest location of the printers across Northeastern campus based on student's location, reducing time and effort by 25%.
- Compiled **comprehensive project documents** such as work breakdown structure, project schedule, RACI matrix, risk matrix, SWOT analysis, resource allocation, financial budget, meeting minutes.

Value Stream Mapping of Paper Manufacturing:

- Identified and eliminated **non-value-added tasks** in the paper manufacturing process, resulting in a 28% decrease in **cycle time** and improved overall efficiency.
- Mapped the factory floor and accomplished a future state map to optimize the plant layout, leading to a 10% increase in productivity and a 20% improvement in production efficiency.

LEADERSHIP AND VOLUNTEER EXPERIENCE

Protoy First Generation, Kolkata, WB, India

Jan 2016 to Jul 2022

- Empowered 1500+ first-generation school students through academic assistance, mentorship, and resource connection, fostering academic and personal growth.